

DATABASE NORMALIZATION PROCESS

A Step-by-Step Beginner's Guide

From an unnormalized report to Third Normal Form (3NF)

What normalization does

Normalization evaluates and corrects table structures so that each fact is stored in an appropriate place. It reduces repeated data and prevents **insertion, update and deletion anomalies**.

Why repeated data causes problems

Update anomaly

A Systems Analyst rate appears in several rows. Missing one row during an update creates conflicting rates.

Insertion anomaly

A new job class cannot be recorded cleanly if the design requires an employee assignment first.

Deletion anomaly

Deleting the last assignment for a project can accidentally remove the only stored project details.

The route from UNF to 3NF

UNF

Repeating groups or report-style nested data remain.

1NF

One value per cell, one row per occurrence, and a key is identified.

2NF

1NF plus no partial dependencies on part of a composite key.

3NF

2NF plus no transitive dependencies between non-key attributes.

Memory aid: First make the data atomic for 1NF. Then check that every non-key attribute depends on the key, the whole key, and nothing but the key. The saying summarizes dependency checks; it is not the definition of 1NF.

Notation used in this guide

Notation	Meaning	Example
Underlined	Primary-key attribute	<u>Proj_Num</u>
Two underlined attributes	Composite primary key; the pair identifies one row	(<u>Proj_Num</u> , <u>Emp_Num</u>)
Italic with *	Foreign key referencing another relation	<i>Job_Class</i> *
A -> B	A functionally determines B	Job_Class -> Chg_Hour

Step 0: Recognise Unnormalized Form (UNF)

UNF rule

A report is unnormalized when it contains a repeating group or a field that holds several values. Blank report cells may visually mean 'same project as above'; that presentation is not a complete relational row.

Report-style source: PROJECT {Proj_Num, Proj_Name, {Emp_Num, Emp_Name, Job_Class, Chg_Hour, Hours}}

Proj_Num	Proj_Name	Emp_Num	Emp_Name	Job_Class	Chg_Hour	Hours
15	Evergreen	101	John News	DB Designer	105.00	19.4
		102	David Senior	Systems Analyst	96.75	23.8
18	Amber Wave	103	June Arbough	Elect. Engineer	84.50	24.6
		104	Anne Ramoras	Systems Analyst	96.75	45.3
		102	David Senior	Systems Analyst	96.75	14.2

Step 1: Convert the repeating group to First Normal Form (1NF)

1NF rule - atomic rows

Create one row for every project-employee assignment, place one value in every cell, and identify a key. Here the pair (Proj_Num, Emp_Num) is the composite primary key.

ASSIGNMENT_1NF (Proj_Num, Emp_Num, Proj_Name, Emp_Name, Job_Class, Chg_Hour, Hours)

Proj_Num	Emp_Num	Proj_Name	Emp_Name	Job_Class	Chg_Hour	Hours
<u>15</u>	<u>101</u>	Evergreen	John News	DB Designer	105.00	19.4
<u>15</u>	<u>102</u>	Evergreen	David Senior	Systems Analyst	96.75	23.8
<u>18</u>	<u>103</u>	Amber Wave	June Arbough	Elect. Engineer	84.50	24.6
<u>18</u>	<u>104</u>	Amber Wave	Anne Ramoras	Systems Analyst	96.75	45.3
<u>18</u>	<u>102</u>	Amber Wave	David Senior	Systems Analyst	96.75	14.2

1NF removes the repeating group, but it does not remove redundancy. Project and employee facts still repeat.

Read the Functional Dependencies Before Decomposing

Full dependency

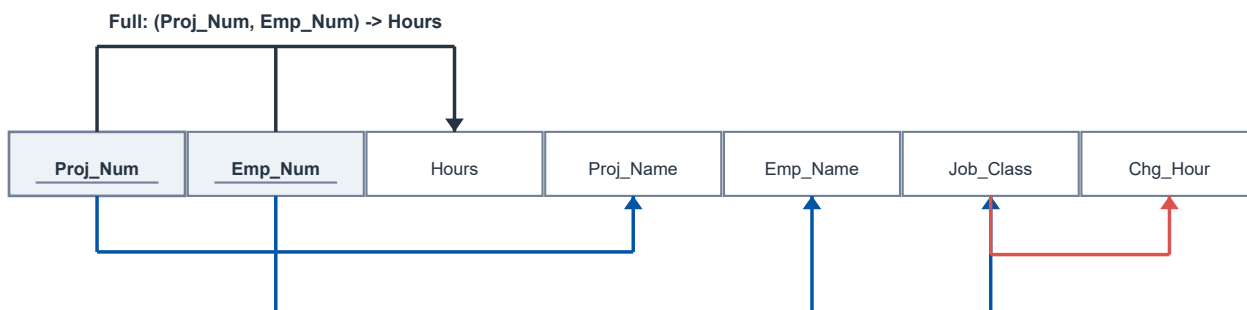
The attribute needs the complete composite key. Example: both project and employee identify Hours.

Partial dependency

A non-key attribute needs only part of a composite key. This prevents 2NF.

Transitive dependency

A non-key attribute depends on another non-key attribute. This prevents 3NF.



Blue = partial dependencies

Red = transitive dependency

Dependency	Classification	What it tells us
(Proj_Num, Emp_Num) -> Hours	Full	Hours belongs to the assignment between one project and one employee.
Proj_Num -> Proj_Name	Partial	Project name belongs in PROJECT.
Emp_Num -> Emp_Name, Job_Class	Partial	Employee facts belong in EMPLOYEE.
Job_Class -> Chg_Hour	Transitive	The charge rate belongs in JOB.

Because Emp_Num determines Job_Class and Job_Class determines Chg_Hour, Emp_Num also determines Chg_Hour indirectly. The direct transitive problem to show is Job_Class -> Chg_Hour.

Normal-form decision

Question	Decision
Are all values atomic and is each assignment a separate row?	Yes - the table is in 1NF.
Do non-key attributes depend on only Proj_Num or only Emp_Num?	Yes - partial dependencies remain, so it is not in 2NF.
Does Chg_Hour depend on the non-key Job_Class?	Yes - a transitive dependency also remains.

Step 2: Convert 1NF to Second Normal Form (2NF)

2NF rule - remove partial dependencies

A relation is in 2NF when it is already in 1NF and every non-key attribute depends on the **whole** primary key. Move facts controlled by only Proj_Num or only Emp_Num into their own relations.

PROJECT (Proj_Num, Proj_Name)

EMPLOYEE (Emp_Num, Emp_Name, Job_Class, Chg_Hour)

ASSIGNMENT (Proj_Num*, Emp_Num*, Hours)

2NF data: the complete EMPLOYEE relation must still contain both Job_Class and Chg_Hour. The transitive dependency is removed only in the next step.

PROJECT

Proj_Num	Proj_Name
15	Evergreen
18	Amber Wave

ASSIGNMENT

Proj_Num*	Emp_Num*	Hours
15	101	19.4
15	102	23.8
18	103	24.6
18	104	45.3
18	102	14.2

EMPLOYEE - complete 2NF relation

Emp_Num	Emp_Name	Job_Class	Chg_Hour
101	John News	DB Designer	105.00
102	David Senior	Systems Analyst	96.75
103	June Arbough	Elect. Engineer	84.50
104	Anne Ramoras	Systems Analyst	96.75

Why this is not yet 3NF: EMPLOYEE still stores Chg_Hour, but the rate is determined by Job_Class. That is the transitive dependency Job_Class -> Chg_Hour.

Step 3: Convert 2NF to Third Normal Form (3NF)

3NF rule - remove transitive dependencies

A relation is in 3NF when it is already in 2NF and no non-key attribute depends on another non-key attribute. Create JOB for Job_Class -> Chg_Hour, then keep Job_Class in EMPLOYEE as a foreign key.

PROJECT (Proj_Num, Proj_Name)

JOB (Job_Class, Chg_Hour)

EMPLOYEE (Emp_Num, Emp_Name, Job_Class*)

ASSIGNMENT (Proj_Num*, Emp_Num*, Hours)

JOB - new in 3NF

Job_Class	Chg_Hour
DB Designer	105.00
Systems Analyst	96.75
Elect. Engineer	84.50

EMPLOYEE - updated

Emp_Num	Emp_Name	Job_Class*
101	John News	DB Designer
102	David Senior	Systems Analyst
103	June Arbough	Elect. Engineer
104	Anne Ramoras	Systems Analyst

Verify the final design

Relation	Key controls	3NF check
PROJECT	Proj_Num -> Proj_Name	Project fact stored once
JOB	Job_Class -> Chg_Hour	Rate stored once per job class
EMPLOYEE	Emp_Num -> Emp_Name, Job_Class	Every non-key fact describes one employee
ASSIGNMENT	(Proj_Num, Emp_Num) -> Hours	Hours depends on the complete composite key

The design preserves the original facts: join ASSIGNMENT to PROJECT and EMPLOYEE, then join EMPLOYEE to JOB. The same report can be reconstructed without storing project names, employee names or rates repeatedly.

Exam Answering Strategy

Common ICT450 question pattern

Past papers commonly ask students to: **(1)** state the current normal form with reasons, **(2)** write the relational schema and identify keys/dependencies, and **(3)** normalize through 3NF while showing every step. Always follow the exact wording and requested format in the paper.

1. Diagnose

Identify repeating groups, non-atomic values and the current normal form. Give evidence from the data.

2. Identify

Propose the key, then write full, partial and transitive dependencies using arrows.

3. Transform

Write the complete 1NF, 2NF and 3NF schemas. Never jump directly to the final tables.

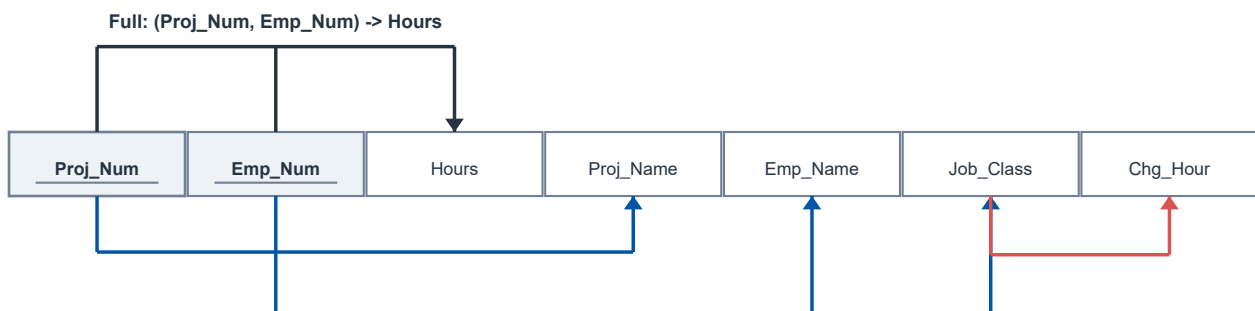
4. Label

Underline every PK and mark every FK consistently. Preserve the associative relation.

Model response for the project-employee example

Prompt	Concise answer
State the original form	UNF because a project record contains a repeating employee group. The blank report cells represent inherited project values, not complete relational rows.
Propose the 1NF key	Composite key: (Proj_Num, Emp_Num). One project can have many employees and one employee can work on many projects.
State the dependencies	Full: (Proj_Num, Emp_Num) -> Hours. Partial: Proj_Num -> Proj_Name; Emp_Num -> Emp_Name, Job_Class. Transitive: Job_Class -> Chg_Hour.
Normalize	Show ASSIGNMENT_1NF, then PROJECT + EMPLOYEE + ASSIGNMENT for 2NF, then add JOB and retain Job_Class* in EMPLOYEE for 3NF.

How to present a dependency diagram



Blue = partial dependencies

Red = transitive dependency

A dependency diagram must show determinants and arrow targets clearly. Do not draw one long arrow and expect the examiner to guess which attributes it controls.

Complete Worked Answer: Show Every Normal Form

1NF - remove the repeating group and identify the composite key

ASSIGNMENT_1NF (Proj_Num, Emp_Num, Proj_Name, Emp_Name, Job_Class, Chg_Hour, Hours)

2NF - remove partial dependencies

PROJECT (Proj_Num, Proj_Name)

EMPLOYEE (Emp_Num, Emp_Name, Job_Class, Chg_Hour)

ASSIGNMENT (Proj_Num*, Emp_Num*, Hours)

3NF - remove the transitive dependency

PROJECT (Proj_Num, Proj_Name)

JOB (Job_Class, Chg_Hour)

EMPLOYEE (Emp_Num, Emp_Name, *Job_Class**)

ASSIGNMENT (Proj_Num*, Emp_Num*, Hours)

Common mistakes to avoid

Mistake	Correction
Calling 1NF 'the key' only	State the real rule: atomic values, no repeating groups and a key identified.
Underlining Proj_Num and Emp_Num without explaining the pair	State explicitly that the two attributes form one composite primary key.
Removing Chg_Hour during the 2NF step	Keep it in EMPLOYEE for 2NF; move it to JOB only when resolving the transitive dependency for 3NF.
Dropping Job_Class from EMPLOYEE in 3NF	Retain Job_Class as the foreign key that connects EMPLOYEE to JOB.
Jumping from UNF directly to 3NF	Show 1NF, 2NF and 3NF in order when the question asks for all steps.
Adding surrogate keys without explanation	Use the natural/composite key required by the dependency analysis unless the question authorizes a redesign.

Final self-check: Can every original fact be reconstructed? Is each primary key clear? Are foreign keys retained? Does every non-key attribute depend on the key, the whole key, and nothing but the key?